

Can the Muon Background Fake a Dark Matter Signal?

Extending DM-Ice17's Annual-Modulation Systematics into the WIMP Signal Region

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1. Objective, Significance, and Implications

For more than two decades the DAMA/LIBRA experiment in Italy has reported that its low-energy event rate rises and falls once a year, peaking near June 2, and has interpreted this as the Earth's orbit sweeping through a galactic halo of dark matter particles. No other experiment has reproduced it, and the central objection is mundane: laboratories also have seasons. Temperature, radon, and cosmic-ray muon flux all vary annually, and any of them could imprint a yearly cycle on a sodium-iodide detector. Distinguishing a galactic signal from an environmental one therefore comes down to *phase*—a dark matter signal keeps its June maximum everywhere on Earth, while most terrestrial effects follow the local seasons and invert in the southern hemisphere.

DM-Ice17 was built to make exactly that comparison. It is a pair of 8.5 kg thallium-doped NaI crystals frozen into the Antarctic ice at 2450 m depth, at the bottom of two IceCube Neutrino Observatory strings, and it is the only NaI dark matter detector ever operated inside a cubic-kilometre muon telescope—which means its dominant seasonal background can be tagged independently, event by event, by the surrounding array.

Over the past summer I established that the DM-Ice muon *flux* modulates by $10.2\% \pm 1.6\%$ and peaks in late January (day-of-year 30 ± 9 , 6.2σ), 13.4σ away from the DAMA phase, and I recovered a muon selection for the 2015–2020 high-gain era that no published analysis had ever reached. The flux, however, is not what a dark matter search actually sees. What enters the 2–6 keV electron-equivalent (keVee) signal region is the *delayed* tail of muon energy deposition—scintillator afterglow and phosphorescence with its own time structure—plus the ordinary low-energy radioactive background. My objective this term is to measure that directly: **does the low-energy background inside the WIMP region of interest modulate, at what amplitude and phase, how much of it is muon-induced, and how much of that the IceCube veto removes?**

This matters for three reasons. First, it converts a statement about muons into a statement

about a dark matter search: an ROI modulation amplitude and phase is the quantity a DAMA test is actually made of, and it has never been measured for a southern-hemisphere NaI detector. Second, the muon-induced leakage fraction and the corresponding veto rejection efficiency are the first such numbers for a NaI target inside an active muon detector, and they are a concrete design input for future in-ice or veto-instrumented NaI deployments in the DM-Ice→COSINE-100 program. Third, the underlying dataset gains lasting value: every published DM-Ice muon analysis stops in January 2015, two months after the photomultiplier gain was raised roughly tenfold, so the majority of the exposure has never been characterized. Completing that characterization is a publishable result in its own right and the prerequisite for any use of the 2015–2020 data.

2. Preparation Completed

This is a continuation, not a new start, and the load-bearing infrastructure already exists and is validated. I have built a stitched waveform-summary extraction over the full 2011–2022 DM-Ice run (2.38 billion events reduced to 136 columnar files), which solves the problem that high-gain signals span three digitizer channels with no single channel covering the full gamma-to-muon range. I built the first real DM-Ice live-time database in the group’s code base—per-run active spans binned by day of year across 95,174 runs, yielding 1635 low-gain and 2328 high-gain live-days per detector—replacing the calendar-day approximation every prior script used; live-time normalization is the load-bearing systematic for any modulation fit. I reproduced the published pulse-shape muon populations as a baseline (2.89–2.92 muons/crystal/day), then diagnosed and fixed a $751\times$ over-count when that selection was transferred to the high-gain era, reaching 2.918 ± 0.038 muons/crystal/day at efficiency 0.949 and purity 0.933—statistically indistinguishable from the low-gain benchmark at 0.03σ —using a discriminant never previously applied to these data. I also hold a frozen 218-event IceCube-coincidence muon sample and a low-gain ADC-to-keV calibration validated by the 600 keV thallium/bismuth line. What remains is a well-scoped extension of a working pipeline.

3. Detailed Research Plan

Work Package 1 — High-gain low-energy calibration (the current blocker). In the high-gain era, low-energy events are digitized in the high-gain ATWD0 channel, but my existing summary columns carry the low-gain ATWD2 channel, which is empty for $\sim 97\%$ of those events; the 2–6 keVee region is therefore simply absent from the current reduction. I will complete the ATWD0-equivalent extraction (already written and tested on a 20-file-per-month subsample for 2013 and 2017), run it over 2015–2020 on IceCube’s Cobalt/Condor infrastructure, and fit per-era, per-detector energy scales anchored on the 600 keV Tl/Bi and 1460 keV ^{40}K gamma lines. *Deliverable:* a trustworthy keV scale and a defensible ROI definition in both gain eras. *Validation:* the same procedure applied to 2013 must reproduce the independently verified low-gain calibration.

Work Package 2 — ROI modulation fit. On live-time-normalized day-of-year histograms of non-muon-tagged events with 2–6 keVee deposits, I will fit $R(t) = R_0[1 + A \cos(2\pi(t - t_0)/365)]$ separately per detector and per gain era, since selection thresholds differ. Fixing all cuts before examining the phase protects against tuning toward an answer, and the two eras—different gain, different readout channel, different years—serve as each other’s cross-check. *Deliverable:* A and t_0 with uncertainties, compared against the DAMA phase (day 152) and the known austral-summer in-ice muon maximum (day 15–30); a null result is reported as an amplitude limit.

Work Package 3 — Afterglow leakage and veto efficiency. I will measure the rate of ROI events in delayed windows following each tagged muon, from microseconds to seconds, against randomized control windows drawn from the same runs, which isolates the muon-correlated excess and its decay time. Combining that leakage fraction with the IceCube-coincident sample gives the veto rejection efficiency for muon-induced ROI events and the dead-time cost of applying it—the number that closes the loop from “muons modulate” to “muons contribute $X\%$ of the ROI modulation, of which the veto removes $Y\%$.”

Work Package 4 — Two open systematics. The 2015 high-gain Det-1 muon rate sits 12.5σ below 2016–2020 and is demonstrably not a selection artifact; I will test the leading hypothesis—that per-run live-time is overestimated because it cannot resolve intra-run dead time—by comparing per-run

event-rate and span distributions against later years. Separately, an unidentified high-amplitude population in high-gain Det-1 (bimodal decay parameter at 173 and 182 ns, and confirmed not to be digitizer saturation) is a ^{210}Po alpha candidate that becomes testable once WP1's calibration exists.

Risks and fallbacks. If the full re-extraction proves too slow on shared storage, I will restrict WP2 to Det-2 and 2016–2020, a subsample already shown to be clean and year-stable, which preserves the physics result at reduced statistics. If the 2–6 keVee band turns out to be dominated by the trigger turn-on, I will raise the band to 6–20 keVee and report the phase there plus a limit below. All selections live in a single canonical module so that no two scripts can disagree about a cut, and IceCube coincidence remains available throughout as an independent hard label.

4. Time Frame (September–December 2026)

I plan on 10–12 hours per week; long extractions run unattended on Cobalt/NPX, so compute time does not compete with coursework.

Weeks 1–3 (Sep 8–26): finalize the ATWD0-equivalent extraction and validate it on 2013 low-gain data, where the correct answer is already known.

Weeks 4–6 (Sep 29–Oct 17): full 2015–2020 extraction on Condor; gamma-line calibration fits per era and detector. *Milestone 1: high-gain keV scale and ROI definition, reviewed with my mentor.*

Weeks 7–9 (Oct 20–Nov 7): WP2 modulation fits in both eras; present results at group meeting. *Milestone 2: ROI amplitude and phase figure.*

Weeks 10–12 (Nov 10–28): WP3 afterglow correlation and veto rejection efficiency.

Weeks 13–15 (Dec 1–19): WP4 systematics; write an internal DM-Ice/IceCube analysis note, draft a conference abstract, and submit the final HRCP report.

5. Faculty Involvement and Support Network

My faculty mentor is Prof. Carlos A. Argüelles Delgado (Department of Physics), who leads Harvard's IceCube group and has supervised this project throughout. He sets the physics scope and priorities, reviews results at each milestone, and has previously required me to freeze and defend a completed analysis in writing before moving on—a practice that produced the wrap-up memo

underpinning this proposal. He supervises through weekly group meetings and regular one-on-one meetings, provides the IceCube Collaboration channel through which this work receives internal review, and provides the computing access the project depends on: IceCube’s Cobalt and NPX systems and the group’s GPU allocation on the FAS research cluster.

Day-to-day technical supervision comes from the group’s postdoctoral researcher, Felix Yu, who has defined several of the concrete tasks I have completed—including the simulation energy-floor characterization and the IceCube MuonFilter overlap check—and who advises on reconstruction and machine-learning methods. Beyond the group, the IceCube Collaboration’s internal review process provides external scrutiny before any result is published, and the DM-Ice and COSINE-100 literature, especially A. J. F. Hubbard’s 2015 thesis, provides the published pulse-shape baseline against which every step of my selection is validated. I also maintain written session handoffs across the three machines this analysis runs on, which makes my work auditable by my mentor and reproducible by whoever inherits it.

Selected references. A. J. F. Hubbard, *Muon-Induced Backgrounds in the DM-Ice17 NaI(Tl) Dark Matter Detector*, PhD thesis, University of Wisconsin–Madison (2015). · J. Cherwinka *et al.* (DM-Ice), “First data from DM-Ice17,” *Phys. Rev. D* **90**, 092005 (2014). · R. Bernabei *et al.* (DAMA/LIBRA), “First model-independent results from DAMA/LIBRA-phase2,” *Universe* **4**, 116 (2018). · G. Adhikari *et al.* (COSINE-100), “An experiment to search for dark-matter interactions using sodium iodide detectors,” *Nature* **564**, 83 (2018). · S. Tilav *et al.* (IceCube), “Atmospheric variations as observed by IceCube,” arXiv:1001.0776 (2010).